AMOUNT OF AIR BLOWN FROM VENTS DOES NOT CHANGE [FULL-AUTO AIR CONDITIONER]

id0703c1700400

DESCRIPTION	• Malfunction in blower system
POSSIBLE CAUSE	• HEATER 40 A fuse malfunction • Blower unit malfunction • Open or short circuit in wiring harness between climate control unit and blower relay • Blower relay malfunction • Blower motor malfunction • Blower fan controller malfunction • Climate control unit malfunction

- When performing an asterisked (*) troubleshooting inspection, shake the wiring harness and connectors while performing the inspection to discover whether poor contact points are the cause of any intermittent malfunctions. If there is a problem, inspect make sure connectors, terminals and wiring harnesses are connected correctly and undamaged.

Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT FUSE** • Remove the HEATER 40 A fuse. • Is the fuse normal?	Yes	Go to the next step.
		No	**If the fuse is melted:** Repair or replace wiring harness for short to ground and replace the fuse. **If the fuse is deterioration:** Replace the fuse. Then go to Step 15.
2	**INSPECT TO SEE WHETHER MALFUNCTION IS IN BLOWER UNIT OR ELSEWHERE** • Switch the ignition ON (engine off or on). • Turn the fan dial clockwise. • Recirculate air inside the vehicle. • Does the blower motor rotate smoothly?	Yes	Go to the next step.
		No	Go to Step 4.
3	**INSPECT BLOWER UNIT INTAKE VENT** • Is blower unit intake vent restricted?	Yes	Remove obstruction, then go to Step 15.
		No	Inspect if there are any obstruction in the A/C unit passage, then go to Step 15.
4*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN BLOWER RELAY SYSTEM OR BLOWER FAN CONTROLLER SYSTEM** • Switch the ignition ON (engine off or on). • Turn the fan dial clockwise. • Measure the voltage at the following blower motor terminal. — Terminal B (blower motor operation signal) • Is voltage **approx. 12 V**?	Yes	Go to Step 8.
		No	Go to the next step.
5*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY BETWEEN FUSE BLOCK AND BLOWER RELAY) OR ELSEWHERE** • Measure the voltage at the following blower relay terminals. — Terminal A (B+ signal) — Terminal B (B+ signal) • Is the voltage **approx. 12 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between blower relay terminal and HEATER 40 A fuse. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 15.

Step	Inspection		Action
6*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY BETWEEN BLOWER RELAY AND GROUND) OR ELSEWHERE** • Switch the ignition ON (engine on or off). • Measure the voltage at the following blower relay terminal. — Terminal B (ground) • Is the voltage **approx. 0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between blower relay terminal and front control module (BCM) terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 15.
7*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY BETWEEN BLOWER RELAY AND BLOWER FAN CONTROLLER) OR BLOWER RELAY** • Switch the ignition ON (engine off or on). • Measure the voltage at the following blower relay terminal. — Terminal C (blower motor operation signal) • Is the voltage **approx. 12 V**?	Yes	Repair the wiring harness for lack of continuity between the blower relay and blower fan controller, then go to Step 15.
		No	Replace the blower relay, then go to Step 15. (See RELAY LOCATION.)
8*	**INSPECT BLOWER MOTOR** • Inspect the blower motor. (See BLOWER MOTOR INSPECTION.) • Is the blower motor normal?	Yes	Go to the next step.
		No	Replace the blower motor, then go to Step 15. (See BLOWER MOTOR REMOVAL/INSTALLATION.)
9*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY BETWEEN BLOWER MOTOR AND BLOWER FAN CONTROLLER) OR ELSEWHERE** • Measure the voltage at the following terminal of blower fan controller. — Terminal E (blower motor operation signal) • Is voltage **approx. 12 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between blower motor terminal and blower fan controller terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 15.
10*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY BETWEEN BLOWER FAN CONTROLLER AND GROUND) OR ELSEWHERE** • Measure the voltage at the following blower fan controller terminal. — Terminal C (ground) • Is the voltage **approx. 0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between blower fan controller terminal and ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 15.
11	**INSPECT BLOWER UNIT** • Inspect the fan in blower unit. — Is the fan free of interference with the blower unit case? — Is the fan free of foreign matter and obstructions? • Is the fan normal?	Yes	Go to the next step.
		No	Remove obstruction, repair or replace the fan and blower unit case, then go to Step 15.
12*	**INSPECT BLOWER FAN CONTROLLER** • Inspect the blower fan controller. (See BLOWER FAN CONTROLLER INSPECTION.) • Is the blower fan controller normal?	Yes	Go to the next step.
		No	Replace the blower fan controller, then go to Step 15. (See BLOWER FAN CONTROLLER REMOVAL/INSTALLATION.)

Step	Inspection		Action
13*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY BETWEEN BLOWER FAN CONTROLLER AND CLIMATE CONTROL UNIT) OR ELSEWHERE** • Switch the ignition off. • Disconnect climate control unit connector. • Inspect for continuity at the following terminals between the blower fan controller and climate control unit. — Terminal B—Q (blower motor speed control signal) • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between blower fan controller terminal and climate control unit terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 15.
14*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN CLIMATE CONTROL UNIT OR WIRING HARNESS (SHORT TO GROUND IN WIRING HARNESS BETWEEN BLOWER FAN CONTROLLER AND CLIMATE CONTROL UNIT)** • Inspect for continuity at the following terminal between the blower fan controller and ground. — Terminal B (blower motor control signal)—ground • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between blower fan controller terminal and ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to the next step.
		No	Replace the climate control unit, then go to the next step. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].)
15	**VERIFY THAT MALFUNCTION SYMPTOM DOES NOT RECUR AFTER REPAIR** • Is air discharged from vent?	Yes	Troubleshooting completed. Explain repairs to customer.
		No	Recheck malfunction symptoms, then repeat from Step 1 if the malfunction recurs.